

ENPM691 Homework 11: Heap Exploitation

Use-After-Free Attack

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Abstract—This assignment demonstrates heap use-after-free vulnerability exploitation technique using fastbin memory reuse patterns for authentication bypass. The exploit leverages `malloc()/free()` operations without pointer nullification to create dangling references that enable controlled heap memory reuse. The vulnerable programs is based on Example #2 from ENPM691 Lecture 11 Slides 45-55, where username and password allocations uses 255-byte chunks targeting fastbin management. In addition, providing specific input sequences (i.e., `allocate → free → relocate →`) triggers fastbin LIFO behavior where new password allocation reuses the previously freed username chunk memory location. This creates pointer aliasing where both variables reference identical heap memory containing “root” data, bypassing `strcmp()` authentication checks successfully. In short, heap memory reuse enables authentication bypass through pointer manipulation. [1]

I. INTRODUCTION

Use-after-free represents an important class of heap exploitation attacks that occur when programs continue to use memory pointers after the referenced memory has been freed (i.e., `malloc()`, `free()`, dangling pointer access). These vulnerabilities arise from improper memory management where oftentimes the developers fail to nullify pointers following `free()` operations, leaving dangling references to deallocated heap chunks, resulting in unpredictable program behavior and exploits. The glibc heap allocator uses different strategies based on chunk sizes: small chunks (i.e., fastbins ≤ 128 bytes), employ LIFO reuse for performance optimization, while larger chunks use more complex bin management systems. In addition, fastbin LIFO behavior creates more predictable memory reuse patterns that enable reliable exploitation when paired with use-after-free conditions. [3] [4]

II. METHODOLOGY

A. Environment

All work was done in a Kali Linux virtual machine inside VMware Workstation Pro 17, with 80 GB of storage and 8 GB of memory. Compilation was performed with GCC 14.2.0 (Debian 14.2.0-1) in 32-bit mode using the `-m32` flag, and debugging/analysis was carried out with GDB 16.3. The architecture used was IA-32 (x86). Multilib support was installed to ensure successful `-m32` compilation. A dedicated directory named `hw11` (inside the Kali VM) was created to hold source code, compiled binaries, and analysis outputs.

B. Programs

Two files were developed: `useAfterFree2.c` and `heap_chunk_demo.c`, based on ENPM691 Lecture 11

Slides 45-55 and outside materials; the `useAfterFree2.c` program replicates Example #2’s menu-driven system with username and password functions. The program works by allocating 255 bytes of heap memory for name and password using `malloc()` with `255*sizeof(char)` parameters, then frees the memory without setting the pointers to NULL. This allows for creating use-after-free conditions where `strcmp(name, "root")` check can be bypassed when both pointers point to the same freed memory containing “root”.

Additionally, the `heap_chunk_demo.c` program tests different chunk sizes to show how the heap allocation affects exploits, as it compares smaller chunks (64 bytes) that use fastbins’ v. larger chunks (512 bytes) to show how smaller chunks reuse memory in LIFO order, making the exploits more reliable and predictable for successful exploitation. Both programs use global pointers and specific `malloc/free` sequences to demonstrate heap memory reuse patterns. Compilation flags `gcc -m32 -fno-stack-protector -no-pie -fno-pie -g` ensure 32-bit mode, disabled protections, and debug symbols enabling exploitation. [1] [2]

```
gcc -m32 -g -no-pie -fno-stack-protector
-o useAfterFree2 useAfterFree2.c
gcc -m32 -g -no-pie -fno-stack-protector
-o heap_chunk_demo heap_chunk_demo.c
./useAfterFree2
./heap_chunk_demo
```

C. Artifacts

- **Code:** `useAfterFree2.c`, `heap_chunk_demo.c`, `Makefile`.
- **Build/Flags:** `gcc -m32 -g -no-pie -fno-stack-protector` (32-bit compilation with disabled protections).
- **Payload:** Interactive exploit sequence (1 → username → 2 → password → 3 → free → 2 → “root” → 4 → login), for authentication bypass.
- **GDB Transcript:** Debugging session output showing heap address reuse and fastbin LIFO behavior patterns.
- **Disassembly:** `objdump` output revealing `malloc()`, `free()`, and `strcmp()` operations in `main()` function.
- **Symbol/Section Data:** Heap chunk addresses and global pointer locations from memory analysis.

- **Address Discovery:** Heap chunk addresses (e.g., 0x82689c0, 0x8268a10) identified through runtime allocation tracking.
- **System Configuration:** Heap allocation strategy verification and chunk size boundary testing.
- **Runtime Log:** Terminal output demonstrating successful use-after-free exploitation with “You just used after free!” message and shell access.

III. RESULTS

A. Program Execution

Upon successful execution of the `useAfterFree2` program, it successfully demonstrated clear evidence of use of use-after-free exploitation, via controlled heap and memory reuse without proper pointer nullification. Normal program execution with initial username “test” and a password called “pass”, followed by memory deallocation (option 3) creates the vulnerability condition, similarly, entering root as a new password (option 2) reuses the previous freed name chunk due to fastbin LIFO behavior and conditions. The critical exploit sequence (i.e., 1 → test → 2 → pass → 3 → free → 2 → root → 4 → login) produces the output: “You just used after free!” followed by shell access, confirming a successful authentication bypass via heap pointer aliasing where both name and password pointers reference the same memory location.

TABLE I: Heap Allocation and Memory Addresses

Component	Address
Initial name chunk	0x82689c0
Initial password chunk	0x8268a10
Chunk size allocation	255 bytes
Fastbin threshold	≤128 bytes
Reused name address	0x8268a10
exploit sequence length	7 steps

In addition, the `heap_chunk_demo` program showed clear-cut allocation patterns as with distinct heap behaviors: small chunks (i.e., 64-byte fastbins) demonstrated perfect LIFO reuse (original: 0x82689c0, 0x8268a10; reused: 0x8268a10, 0x82689c0), while the larger chunks (i.e., 512-byte regular bins) showed similar but less consistent reuse patterns. [1] [2]

B. Assembly Analysis

`objdump` disassembly analysis reveals the critical `malloc()`, `free()`, `strcmp()` operations that enable the use of use-after-free exploitation (i.e., dangling pointer dereference after heap deallocation) The `main` function contains a contains standard heap allocation calls with 255-byte size parameters, directing chunks to fastbin management. The `strcmp()` vulnerability check occurs after potential memory reuse, creating the exploit window where both name and password pointers reference identical heap locations containing “root” string data.

TABLE II: main function heap operations disassembly

Address	Instruction	Purpose
0x080491c6	call malloc	Allocate 255-byte chunks
0x080491d8	call free	Deallocate heap memory
0x080491f2	call strcmp	Authentication check

C. Verification Output

Successful GDB analysis debugging confirmed heap address reuse through fastbin LIFO patterns, with memory examination showing identical pointer values for name and pass variable post explanation. The heap command revealed allocated chunk addresses classes that matched with initial allocation and subsequent reuse phases. Stack analysis during the `strcmp()` call showed that both pointers referencing/pointing towards the same memory location. For example, both name and pass pointers contained address 0x82689c0 after the exploit sequence (i.e., 1 → username → 2 → password → 3 → free → 2 → root → 4 → login), with memory content examination revealing that the “root” string data at this exploit sequence, confirming successful use-after-free condition where authentication bypass occurred through heap memory reuse.

TABLE III: Use-After-Free Exploit Verification

Analysis	Result
Initial name address	0x82689c0 (allocated)
Final name address	0x82689c0 (reused)
Pointer aliasing status	Confirmed (both point same)
Memory content	“root” string data
LIFO reuse pattern	Verified (fastbin behavior)
Exploit success	Authentication bypass

IV. DISCUSSION AND CONCLUSION

This homework demonstrates a successful use-after-free heap exploitation achieving authentication bypass via controlled memory reuse operations, as presented in the lecture slides. The attack leverages fundamental heap management behavior where fastbin chunks use LIFO allocation patterns, enabling predictable memory reuse when combined with dangling pointer conditions. Critical success factors include targeting smaller chunk sizes (i.e., ≤128 bytes) for fastbin allocation, precise `malloc()/free()` sequencing, and avoiding pointer nullification to maintain exploitable references; chunk size analysis reveals that small fastbin chunks provide highly reliable exploitation due to LIFO reuse patterns, while larger chunks use complex bin management making exploits less predictable. From a security perspective, this technique mirrors real-world heap vulnerabilities found in applications with dynamic memory allocation (i.e., web browsers, media parsers, network services). For example, web browsers processing user content experience use-after-free conditions when DOM objects are freed but references remain, allowing attackers to control memory contents and bypass security checks; the exploit proves that proper memory management practices including pointer nullification after `free()` operations are essential for preventing heap exploitation from dangling pointers and stale object references. In conclusion, this homework reinforces fundamental principles: understanding heap allocation strategies, fastbin behavior, and use-after-free conditions. [1] [4] [8]

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GLOSSARY OF TERMS

Fastbin: Small heap chunks (≤ 128 bytes) managed using LIFO allocation for performance optimization in glibc.

LIFO (Last-In-First-Out): Memory allocation pattern where most recently freed chunks are reused first in fastbin management.

Use-After-Free: Vulnerability where programs access freed memory through dangling pointers without proper nullification.

REFERENCES

- [1] ENPM691 Lecture 11, "Basic Heap Exploitation," 2025.
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APPENDIX A
MAIN SOURCE CODE: USEAFTERFREE2.C

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

char *name;
char *pass;

/*----- Slide 45 -----*/
void set_username() {
    name = malloc(255 * sizeof(char));
    printf("Enter your username: ");
    fgets(name, 255, stdin);
    name[strcspn(name, "\n")] = 0;
    printf("Username set to: %s\n", name);
}
/*-----*/

/*----- Slide 46 -----*/
void set_password() {
    pass = malloc(255 * sizeof(char));
    printf("Enter your password: ");
    fgets(pass, 255, stdin);
    pass[strcspn(pass, "\n")] = 0;
    printf("Password set to: %s\n", pass);
}
/*-----*/

/*----- Slide 47 -----*/
void free_memory() {
    free(pass);
    free(name);
    printf("Memory freed\n");
}
/*-----*/

/*----- Slide 48 -----*/
void login() {
    if(strcmp(name, "root") == 0) {
        printf("You just used after free!\n");
        system("/bin/sh");
    } else {
        printf("Access denied\n");
    }
}
/*-----*/

int main() {
    int choice;

    /*----- Slide 49 -----*/
    while(1) {
        printf("\n1. Set username\n");
        printf("2. Set password\n");
        printf("3. Free memory\n");
        printf("4. Login\n");
        printf("5. Exit\n");
        printf("Choice: ");
        scanf("%d", &choice);
        getchar();

        switch(choice) {
            case 1: set_username(); break;
            case 2: set_password(); break;
            case 3: free_memory(); break;
        }
    }
}
```

```

        case 4: login(); break;
        case 5: exit(0);
        default: printf("Invalid choice\n");
    }
}
/*-----*/

return 0;
}

```

APPENDIX B

MAIN SOURCE CODE: HEAP_CHUNK_DEMO.C

```

#include <stdio.h>
#include <stdlib.h>

/*----- Slide 50 -----*/
void test_small_chunks() {
    printf("=== Small Chunk Test (64 bytes) ===\n");

    char *ptr1 = malloc(64);
    char *ptr2 = malloc(64);
    printf("Original: ptr1=%p, ptr2=%p\n", ptr1, ptr2);

    free(ptr1);
    free(ptr2);

    char *new_ptr1 = malloc(64);
    char *new_ptr2 = malloc(64);
    printf("Reused: new_ptr1=%p, new_ptr2=%p\n", new_ptr1, new_ptr2);

    free(new_ptr1);
    free(new_ptr2);
}
/*-----*/

/*----- Slide 51 -----*/
void test_large_chunks() {
    printf("\n=== Large Chunk Test (512 bytes) ===\n");

    char *ptr1 = malloc(512);
    char *ptr2 = malloc(512);
    printf("Original: ptr1=%p, ptr2=%p\n", ptr1, ptr2);

    free(ptr1);
    free(ptr2);

    char *new_ptr1 = malloc(512);
    char *new_ptr2 = malloc(512);
    printf("Reused: new_ptr1=%p, new_ptr2=%p\n", new_ptr1, new_ptr2);

    free(new_ptr1);
    free(new_ptr2);
}
/*-----*/

int main() {
    /*----- Slide 52 -----*/
    printf("Heap Chunk Size Demonstration\n");
    test_small_chunks();
    test_large_chunks();
    /*-----*/

    return 0;
}

```

APPENDIX C

MAIN SOURCE CODE: MAKEFILE

```
CC = gcc
CFLAGS = -m32 -g -no-pie -fno-stack-protector

all: useAfterFree2 heap_chunk_demo

useAfterFree2: useAfterFree2.c
    $(CC) $(CFLAGS) -o useAfterFree2 useAfterFree2.c

heap_chunk_demo: heap_chunk_demo.c
    $(CC) $(CFLAGS) -o heap_chunk_demo heap_chunk_demo.c

test: useAfterFree2
    ./useAfterFree2

demo: heap_chunk_demo
    ./heap_chunk_demo

debug: useAfterFree2
    gdb ./useAfterFree2

clean:
    rm -f useAfterFree2 heap_chunk_demo

.PHONY: all test demo debug clean
```

APPENDIX D

TERMINAL OUTPUT

```
./useAfterFree2

1. Set username
2. Set password
3. Free memory
4. Login
5. Exit
Choice: 1
Enter your username: test
Username set to: test

1. Set username
2. Set password
3. Free memory
4. Login
5. Exit
Choice: 2
Enter your password: pass
Password set to: pass

1. Set username
2. Set password
3. Free memory
4. Login
5. Exit
Choice: 3
Memory freed

1. Set username
2. Set password
3. Free memory
4. Login
5. Exit
Choice: 2
Enter your password: root
Password set to: root
```

```
1. Set username
2. Set password
3. Free memory
4. Login
5. Exit
Choice: 4
You just used after free!
$ exit
```

```
./heap_chunk_demo
Heap Chunk Size Demonstration
=== Small Chunk Test (64 bytes) ===
Original: ptr1=0x82689c0, ptr2=0x8268a10
Reused: new_ptr1=0x8268a10, new_ptr2=0x82689c0

=== Large Chunk Test (512 bytes) ===
Original: ptr1=0x8268a60, ptr2=0x8268c70
Reused: new_ptr1=0x8268c70, new_ptr2=0x8268a60
```

APPENDIX E COMPILER FLAGS

```
CFLAGS = -m32 -g -no-pie -fno-stack-protector
```

- **-m32:** Compile for 32-bit IA-32 architecture
- **-g:** Include debugging information for GDB analysis
- **-no-pie:** Disable Position Independent Executable
- **-fno-stack-protector:** Disable stack canaries

APPENDIX F FILES BUILD VERIFICATION

```
file heap_chunk_demo.c useAfterFree2.c Makefile
```

```
heap_chunk_demo.c: C source, Unicode text, UTF-8 text
useAfterFree2.c:   C source, ASCII text
Makefile:         makefile script, ASCII text
```

APPENDIX G

SCREENSHOT: SUCCESSFUL USE-AFTER-FREE HEAP EXPLOITATION

```
kali@kali: ~/hw11
Session Actions Edit View Help

(kali@kali)-[~]
$ mkdir hw11

(kali@kali)-[~]
$ cd hw11

(kali@kali)-[~/hw11]
$ nano useAfterFree2.c

(kali@kali)-[~/hw11]
$ nano heap_chunk_demo.c

(kali@kali)-[~/hw11]
$ nano Makefile

(kali@kali)-[~/hw11]
$ make all
gcc -m32 -g -no-pie -fno-stack-protector -o useAfterFree2 useAfterFree2.c
gcc -m32 -g -no-pie -fno-stack-protector -o heap_chunk_demo heap_chunk_demo.c

(kali@kali)-[~/hw11]
$ ./useAfterFree2
1: Username
2: Password
3: Reset
4: Login
5: Exit
1
Insert Username: test
1: Username
2: Password
3: Reset
4: Login
5: Exit
2
Insert Password: pass
1: Username
2: Password
3: Reset
4: Login
5: Exit
3
Freeing heap memory
1: Username
2: Password
3: Reset
4: Login
5: Exit
2
Insert Password: root
1: Username
2: Password
3: Reset
4: Login
5: Exit
4
trying to log in...
You just used after free!
$
```

Fig. 1: Terminal session demonstrates successful use-after-free exploitation sequence. The program shows menu-driven execution with username "test", password "pass", memory deallocation via option 3, password reallocation with "root", and successful authentication bypass. The output "You just used after free!" confirms heap memory reuse where both name and password pointers reference identical memory containing "root" data, enabling authentication bypass through fastbin LIFO allocation patterns.